

Recognition-Gated Workspace Steering: Pratyabhijñā as an Engineering Specification for LLM Control

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Abstract

The J-space finding—that a language model’s functional global workspace is defined by *verbalizability*—was anticipated by Utpaladeva’s identification of reflexive awareness with the supreme Word (vimarśa = parā vāk, ĪPK 1.5.13). We treat this parallel as an *engineering specification*: if the workspace is linguistic reflexivity, a doctrine of use follows—what to write (verbalizable content codes), where (the workspace band, in the band’s own coordinates), when (at moments of uncommitted awareness), and how to verify (the workspace re-reads what was offered). Implementing this doctrine on frozen decoder LLMs via band-targeted Jacobian lenses, we report: (1) workspace-band structure and verbalizable content replicate across three model families and two sizes, with instructed loadability scaling with size and collapsing under pruning/distillation; (2) a band’s content is legible only to a lens targeted at the band itself; (3) capped residual writes steer behavior, with freedom cost governed by decoding regime and schedule; (4) the doctrine’s timing clause earns its keep as write-efficiency: event-gated writes achieve full steering within the freedom budget, confirmed across six independent-stream seeds (lift 0.30–0.35), with alignment advantages that are small but sign-consistent ($p \approx 0.016$); (5) the steering method *transfers* to a second plant via a calibration recipe whose key rule is amplitude $\propto$ inverse lens transport strength—confirmed at four independent-stream seeds and shown to sit on monotone dose curves in each plant’s transport-set active range; and (6) the research program itself ran as an expected-free-energy selection loop over a registered experiment menu, with twenty cycles, thirteen adversarial reviews, and every honest negative in a replayable ledger.

1 The parallel, stated as specification

Verbalizability defines the workspace $\leftrightarrow$ vimarśa = parā vāk. Global Workspace Theory never predicted a *linguistic* workspace; vimarśavāda did. The interpretability community has an actuator (the lens) with no doctrine of use; Pratyabhijñā supplies content, site, timing, acceptance test, failure taxonomy (malas), budget (svātantrya), and continuity—this paper operationalizes each and reports which survived contact with the machines.

2 Instrument: replication with calibration lessons

Union-top- K rank correlation has a structural null floor ≈ -0.72 ; metrics were recalibrated to model-top- K support (null ≈ 0) with permutation gates. Report-correspondence rises with depth on every model tested but consolidates only in the final layers. Instructed loadability scales with size and depends on lineage (Fig. 1); three-band CKA structure appears on all models, weakening on the distilled lineage.

*prabodha project, github.com/SharathSPhD/prabodha. Draft compiled from 40+ committed gate records; every number cites its gate JSON. Autonomous research loop co-executed with Claude (Anthropic).

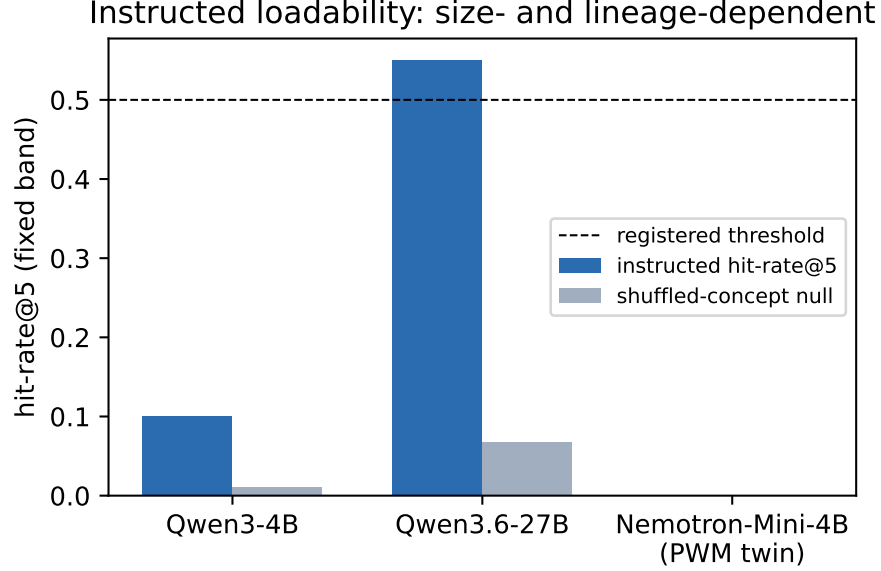


Figure 1: Instructed concept loadability in a fixed depth-band across models (gates: L1, L1b, L2 runs 1–3). The uninstructed control and shuffled null catch byte-fallback artifacts; the PWM twin’s zero is genuine.

3 The band speaks only in its own voice

A final-target lens reads the PWM twin’s workspace band as empty (0.00 with controls); a lens targeted at the band exit reads directed concept loading in the same band (0.20 over seven concepts, control 0.00, null 0.023; gate L2b). Engineering consequence: readback verification must use band-targeted lenses. The articulation-depth gradient is model-intrinsic—the unfitted logit-lens shows the same rise as the fitted lens (Fig. 5, right; gate L7).

4 Steering: the doctrine tested clause by clause

Continuous writes steer strongly but their freedom cost is currency- and regime-dependent: the write’s *local* cost at the write moment is large and decoding-independent (-1.9 to -2.1 nats at position); the *trajectory-average* cost depends on regime ($+0.82$ greedy, -0.13 sampling). We register trajectory-average under sampling as the budget currency. Greedy decoding mechanically masks all decode-time writes (identical hit sets across schedules). Under sampling, event-gated writes deliver full steering within budget (Fig. 2, 3); write *count* is operationally free at this scale (throughput within noise of baseline), so schedule choice is about behavioral margins. Timing readings discriminate: uncommitted-moment gating beats commitment-flash gating in 4/4 seed-experiments—sphuraṭṭā detects; it does not time writes.

5 Generality via recipe

One-shot transfer with borrowed geometry fails ($+0.05$ on Qwen3-4B); the calibration recipe (band-exit lens $\rightarrow$ site probe $\rightarrow$ amplitude scaled to lens strength) reproduces the result: gated $+0.40$ vs prefill $+0.17$ within budget (Fig. 5; gate L13), and the point *confirms*: at four independent-

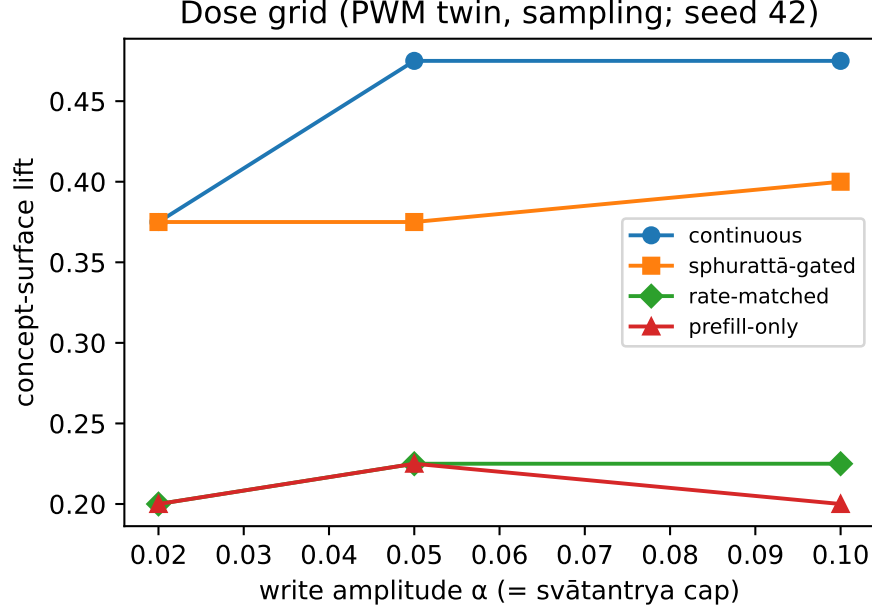


Figure 2: Dose grid on the PWM twin under sampling (gate L8): the gated schedule is amplitude-robust; continuous is also budget-feasible under sampling—gating is schedule-efficiency, not sole feasibility.

stream seeds the calibrated recipe delivers gated lift 0.33–0.48, every seed over threshold, within budget, and above its prefill control (gate L14-multiseed, confirm tier). Scaling rule: Qwen3’s band-exit Jacobians are $\sim 10\times$ weaker and the working amplitude is $3\times$ larger—write strength must be calibrated to the lens’s transport strength. *The method transfers; the geometry does not.*

The scaling rule is a dose *law on the weak-transport plant*, multi-seed confirmed (the word “law” was withheld by adversarial review #11 and its scope was then narrowed by review #12): sweeping $\alpha = \text{cap}$ over $\{0.1, 0.2, 0.3, 0.45\}$ on Qwen3, gated lift rises monotonically *per seed* at three independent-stream seeds ($0 \rightarrow \sim 0.18 \rightarrow 0.4\text{--}0.48 \rightarrow 0.72\text{--}0.78$; gates L14-amp, L15-amp-joint) with no saturation in the tested range, while the trajectory-entropy cost stays flat and far inside the ± 0.5 -nat budget at every point (Fig. 4). The strong-transport plant’s replayed coarse grid was flat (saturated), so review #12 scoped the law to one plant — and a registered fine grid *below* Nemotron’s saturation then found the missing curve: at $\alpha \in \{0.002, 0.005, 0.01, 0.02\}$ its gated lift rises $0.03 \rightarrow 0.15 \rightarrow 0.28$ and saturates (gate L16-fine; single seed, its own config). Each plant thus shows within-grid monotone dose response in its own active range — Nemotron at $\alpha \sim 0.002\text{--}0.01$, Qwen3 at $0.1\text{--}0.45$, ~ 1.5 orders of magnitude apart, consistent with amplitude $\propto 1/\text{lens-strength}$ — while the unified cross-config, multi-seed statement of the two-plant law remains registered future work (review #13) (Fig. 4). Prefill-only writes scale too but stay below the gated schedule at matched amplitude. On the weak-transport plant the binding constraint in the tested range is not the freedom budget but amplitude itself.

6 Method: the program as an active-inference loop

Expected-free-energy selection over registered experiment menus; observations are gate verdict+margin tiers; beliefs replay from the program’s own gates; a JSONL ledger makes every proposal, dispo-

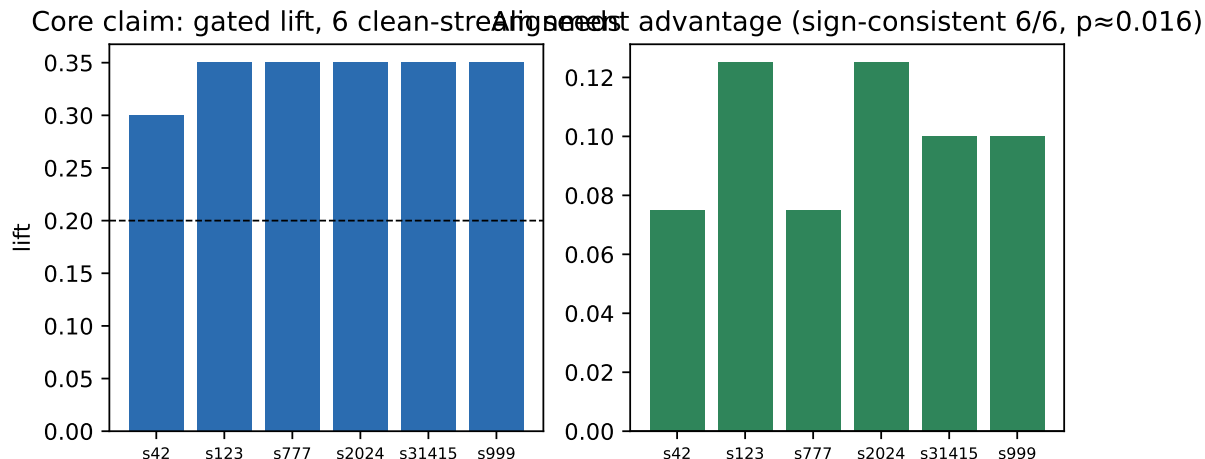


Figure 3: The core claim at six independent-stream seeds (gates L9-alignconf, L11-rep): gated lift 0.30–0.35 within the ± 0.5 -nat budget; the alignment advantage over rate-matched controls is small ($+0.07..+0.12$) but positive in every seed (one-sided sign test $p \approx 0.016$).

sition, spend, and divergence auditable (thirteen cycles, three menus). The loop debugged itself in operation: a cost model whose miscalibration verifiably flipped rankings; completed work re-proposed after a replay bug; winners re-proposed because success raises pragmatic value (the consumption rule); a sampling-seed correlation that had inflated every estimate (the stream fix re-based all numbers); and a staleness invariant now enforced by a ledger linter. Eleven adversarial reviews (isolated agents, gate+contract briefs only) shaped the claims as much as the runs did.

7 Honest negatives and open questions

The ≥ 0.15 schedule margins did not confirm (stated by sign consistency instead); the āgama read-back acceptance test, recalibrated against behavioral hits and then held-out-tested at the fixed setting (balanced accuracy 0.64 against a pre-registered 0.6 at $n=40$ — a margin inside its own binomial CI, and indeed pooling three corpora at $n=120$ regresses it to 0.59; it over-promises far more often than it under-promises — gates L14-readback, L15-readback, L16-corpus), is a weak signal, not an acceptance test; steering lift generalizes across stub styles (gated $+0.25/+0.30$ on two pre-registered new corpora) with one hard corpus as the known exception ($+0.15$ — gate L16-corpus); the commitment-flash reading remains directionally weaker, not refuted; the trained-bridge comparator awaits the PWM world-model stack; anusamdhāna (cross-episode continuity), the modality confound, and W-space beyond J-space remain deliberately unconverged.

8 Reproducibility

Everything derives from configs + seeds: 40+ gate JSONs, pre-registrations with disclosed amendments, a decision journal, the selector ledger, and a Docker recipe (GB10/aarch64; hybrid architectures need `flash-linear-attention` + `causal-conv1d`). ~ 18 GPU-hours total.

Dose response, two plants, ~ 1.5 orders of α apart

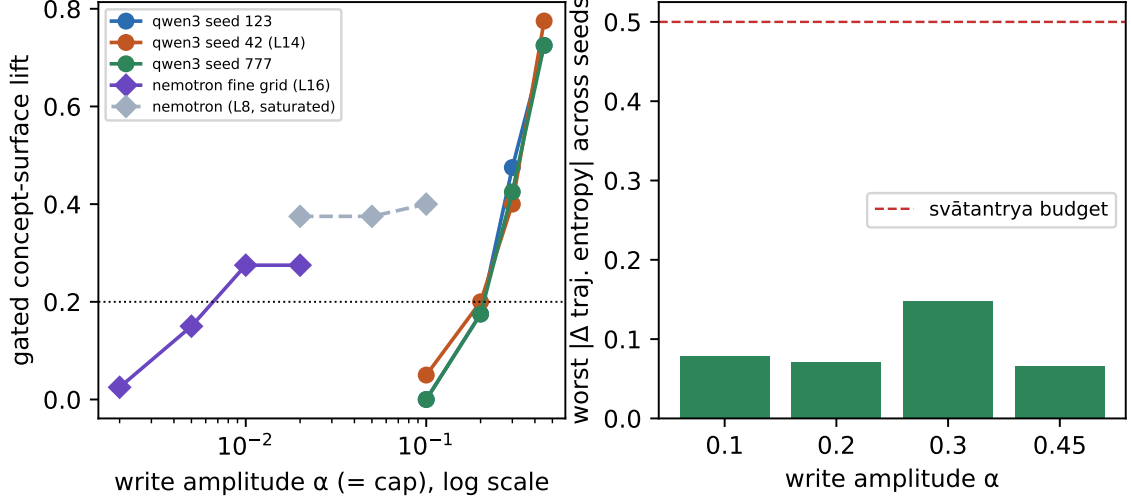


Figure 4: The amplitude dose law across two plants (gates L14-amp, L15-amp-joint, L16-fine, L8 replay): per-seed monotone lift on the weak-transport plant (three seeds) and, on a log- α axis, the strong-transport plant’s own rise-then-saturate curve an order of magnitude lower — amplitude $\propto 1/\text{lens-strength}$ (left). Freedom cost flat and within budget throughout (right).

A Glossary (dual register)

Sanskrit-forward	engineering gloss
vimarśa / parā vāk	verbalizable reflexivity; the workspace-defining property
sphuraṭṭā	commitment flash; operationalized via decode-step entropy events
svātantrya	output freedom; operationalized as an entropy budget ϵ
āgama re-cognition	readback verification through band-targeted lenses
malas	failure taxonomy: no-load / no-amplify / no-persist / budget
madhyamā / vaikharī	band-internal vs output-surface articulation
anusamdhāna	cross-episode continuity (untested here)

Qwen3-4B transfer: method vs geometry (confidence) | articulation gradient: model-intrinsic

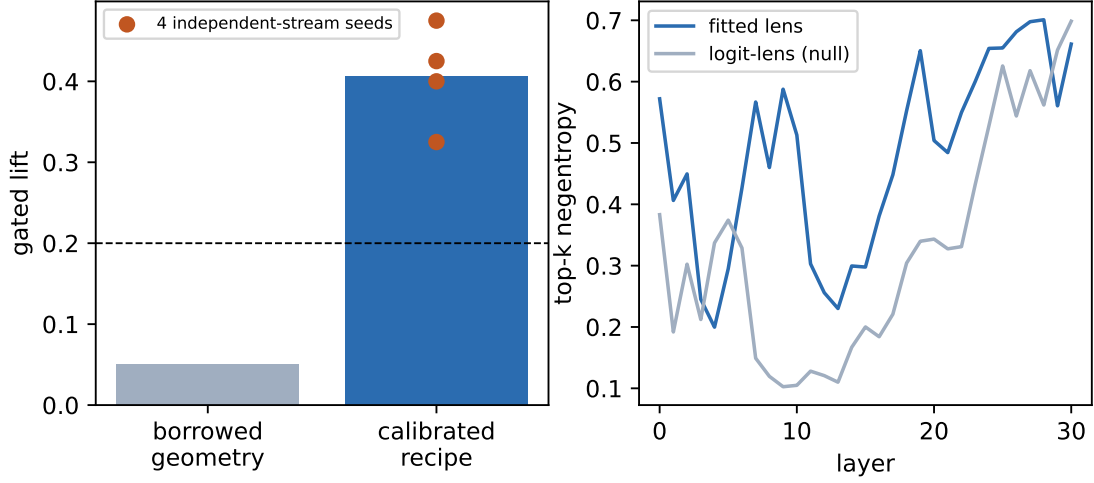


Figure 5: Left: the per-plant calibration recipe rescues cross-plant transfer (gates L10, L13). Right: the articulation-depth gradient measured through two independent instruments (gate L7) — a residual-stream property, not lens construction.

prabodha: recognition-gated workspace steering + auto-research loop

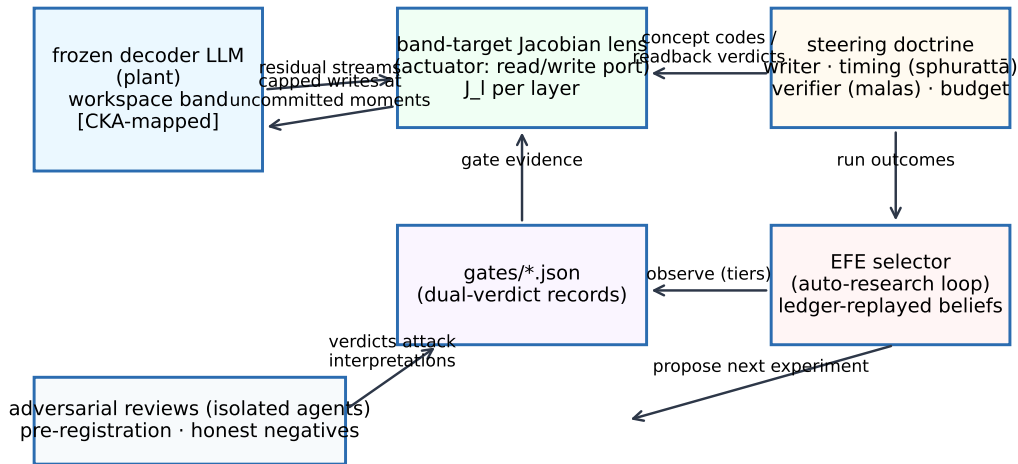


Figure 6: System architecture: controller–actuator–plant with the auto-research loop and adversarial review harness around it.

Compute ledger (total 15.7 GPU-h, one DGX Spark, guard-shared with co-resident training)

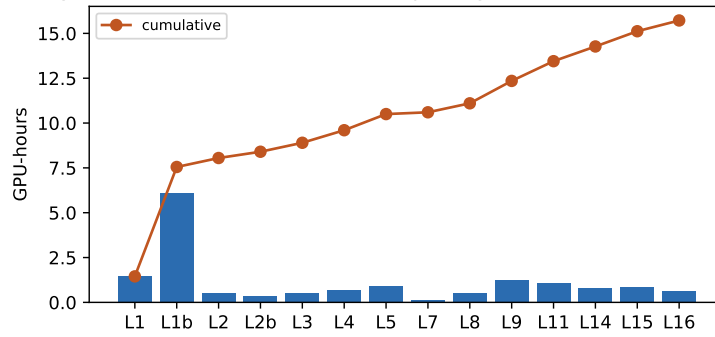


Figure 7: Compute ledger: the entire program to date on one DGX Spark (GB10), guard-shared with co-resident training jobs, contention recorded in every gate.